



TB AUDIO PLUG-INS
Team Impulse Impact

USER MANUAL

TB Inverted Flanger

Inverted Flanger

Phase-inverted modulated delay voices carve hollow, moving notches into the sound, from slow empty-space sweeps to hard metallic motion.

3 inverted delay voices

Zero latency

Output waveform display

6 factory presets



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01 Overview

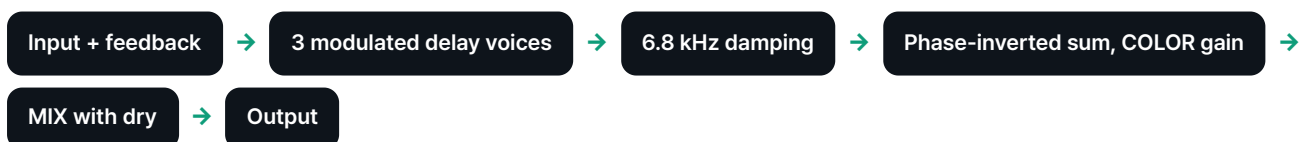
A classic flanger mixes a slowly moving delayed copy with the original. TB Inverted Flanger flips the copies upside down before mixing, so instead of a bright whoosh you get hollow cancellation: deep notches that sweep through the sound.

Three short delay voices, each with its own depth and phase, move together under one LFO. The result ranges from subtle empty-space motion to dramatic sci-fi sweeps on guitars, synths, vocals, and drum loops.

Key features

- Three phase-inverted delay voices with different base delays, depths, and phases
- DEPTH sweeps the delay between roughly 0.1 and 8.4 ms
- FEEDBACK adds resonance, limited internally to keep it under control
- COLOR sets how deep the inverted notches cut, without changing the dry signal
- MIX blends the inverted comb signal with the dry sound
- No added latency; mono input is processed as stereo

Signal flow



About this manual

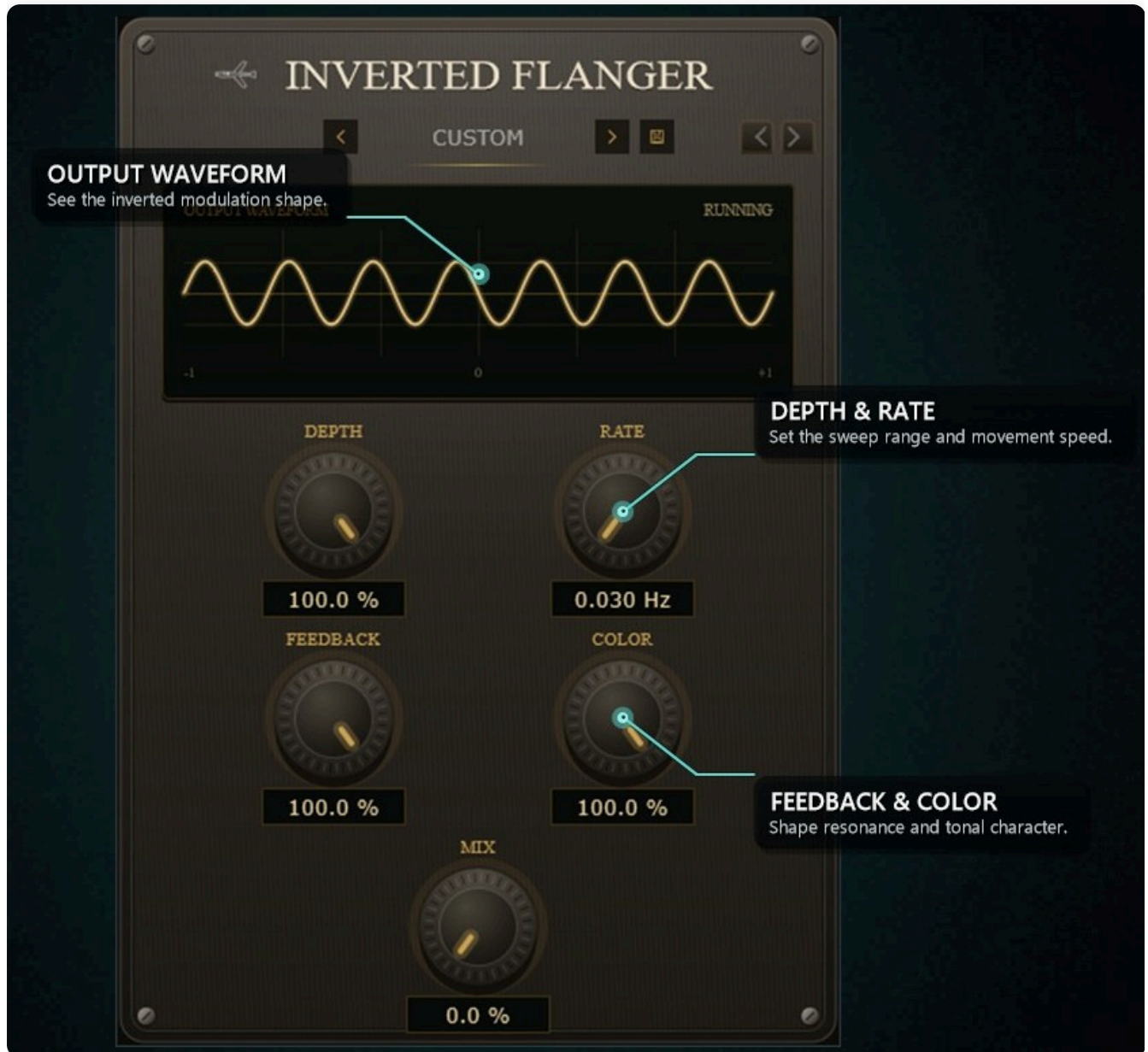
This manual describes TB Inverted Flanger 2.0.0 for Windows (VST3). The current macOS build is version 1.0.2 and may differ from the descriptions here.

02 Quick start

- 1 Insert TB Inverted Flanger on a track. The defaults are **MIX** 52%, **DEPTH** 72%, **RATE** 0.34 Hz, **FEEDBACK** 32%, **COLOR** 68%.
- 2 Play the material and set the speed of the sweep with **RATE**.
- 3 Set how far the notches travel with **DEPTH**.
- 4 Raise **FEEDBACK** for a more resonant, jet-like sweep.

- 5 Raise **COLOR** for deeper, more hollow notches.
- 6 Blend the effect with **MIX**.

03 Interface



TB Inverted Flanger 2.0 interface (portfolio image).

Area	What it does
Preset bar	Previous and next factory preset, the PRESET list (CUSTOM after an edit, LOAD USER PRESET... for your files), Save, Undo, and Redo.
OUTPUT WAVEFORM	Shows the shape of the inverted modulation. RUNNING appears while the modulation runs.
DEPTH , RATE	Range and speed of the sweep.
FEEDBACK , COLOR	Resonance and notch depth.
MIX	Balance between the dry sound and the inverted comb signal.

04 Controls

Control	Range · default	What it does
DEPTH	0 to 100% · 72%	How far the delay is modulated, roughly 0.1 to 8.4 ms. More depth moves the notches across a wider frequency range.
RATE	0.03 to 10 Hz · 0.34 Hz	Speed of the LFO that moves all three delay voices together. Low values give slow sweeps; high values give fast, warbling motion.
FEEDBACK	0 to 100% · 32%	Feeds the output back into the delays for resonance and a stronger jet effect. The engine scales it by 0.9 so it cannot run away.
COLOR	0 to 100% · 68%	Level of the phase-inverted copies, from 0.4 to 1.0 of full gain. Higher values cut deeper notches. The dry signal is not affected.
MIX	0 to 100% · 52%	Blend between the dry sound and the inverted comb signal.

How the inverted voices work

Each of the three voices starts from a different short base delay (0.24, 0.42, and 0.64 ms) and moves with its own depth and phase. Their outputs are damped above about 6.8 kHz and summed upside down, so wherever a voice lines up with the dry signal, that frequency cancels. As the delays move, the cancellation notches sweep through the spectrum.

TB Inverted Flanger has no output knob and adds no latency. Because it works by cancellation, it can make a sound quieter; match levels before comparing with bypass.

05 Presets

Step through the factory presets with the arrows beside the preset name, or open the PRESET list. LOAD USER PRESET... opens a .tbifpreset file, and Save stores the current settings as one. The name shows CUSTOM after you change a preset. The second pair of arrows undoes and redoes changes.

Factory presets: **Empty Orbit**, **Null Corridor**, **Neon Comb**, **Signal Rift**, **Slow Eclipse**, and **10Hz Scanner**.

06 Recipes

Starting points for common jobs. Adjust them by ear.

Hollow guitar sweep

Slow, deep motion on sustained guitar or pads.

Preset Slow Eclipse

RATE 0.1 to 0.3 Hz

DEPTH 70 to 100%

MIX 50 to 70%

1 Raise COLOR until the hollow character is clear.

Jet sweep on drums

A dramatic, resonant whoosh on a drum loop.

FEEDBACK 60 to 90%

RATE 0.1 to 0.5 Hz

MIX 40 to 60%

1 Check that the transients still cut through; lower FEEDBACK if they smear.

Fast scanning texture

Metallic, fast motion for sound design.

Preset 10Hz Scanner

RATE 5 to 10 Hz

DEPTH Low to medium

1 Lower DEPTH if the pitch wobble becomes distracting.

Subtle space on vocals

A light sense of movement without obvious flanging.

Preset Empty Orbit

MIX 20 to 35%

FEEDBACK Low

1 Compare with bypass at matched level.

07 Tips and troubleshooting

Symptom	What to check
The sound got thin or quiet.	That is the cancellation at work. Lower COLOR or MIX, then match levels in your mixer.
The bass disappears.	Use lower DEPTH on bass-heavy sources and raise RATE slowly.
The resonance is too sharp.	Lower FEEDBACK.
The effect is barely audible.	Raise MIX and COLOR, or raise DEPTH.
The mono sum sounds different.	With high FEEDBACK and COLOR, check the result in mono and on the transients.

08 Parameter reference

Panel label	Host parameter	Range	Default
DEPTH	Depth	0 to 100%	72%
RATE	Rate	0.03 to 10 Hz	0.34 Hz

Panel label	Host parameter	Range	Default
FEEDBACK	Feedback	0 to 100%	32%
COLOR	Color	0 to 100%	68%
MIX	Mix	0 to 100%	52%

09 Version history

Version	Changes
2.0.0 · 2026-08-04	The interface was rebuilt as a warm hardware-style rack with a dedicated output waveform display, and DEPTH, RATE, FEEDBACK, COLOR, and MIX were reorganized into a clearer vertical flow.

10 Support and license

Report problems or request features at github.com/TaebeumKim/TB_Audio-Plugins/issues. Include the plug-in version, your DAW, and your operating system.

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